

ML SEC-3

ML PROJECT

*CUSTOMER PURCHASE PREDICTION
&
PERSONALIZED PRODUCT RECOMMENDATION*

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Introduction

The Problem

E-commerce platforms offer thousands of products, making it difficult for customers to discover products that genuinely match their interests.

Our Approach

Our system analyzes real-world e-commerce interaction data—including product views, cart additions, and transactions—to understand customer behavior.

Customer Behavior → Prediction → Personalization

What We Aim to Do

- Predict customer purchase behavior from historical interactions.
 - Identify customers with similar shopping patterns.
 - Use these behavioral insights to support personalized product recommendations.
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Abstract

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- **This project develops a Machine Learning-based system for customer purchase prediction and personalized product recommendation in e-commerce.**
 - **The system analyzes historical customer-product interactions, including product views, add-to-cart actions, and completed transactions. The raw interaction data is explored and transformed into meaningful behavioral features such as interaction frequency, product engagement, cart additions, and purchase activity.**
 - **For purchase prediction, Logistic Regression and HistGradientBoosting are used and compared to identify the better-performing model. The selected model predicts the probability that a customer will purchase a product within 7 days based on their previous behavior. These purchase probabilities are then used to rank products and generate personalized recommendations.**
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Existing System

DATA COLLECTION

The system collects user, product, rating, purchase, and search data from an e-commerce platform to understand user preferences and interactions.

MATRIX FACTORIZATION

Techniques such as SVD/ALS are used to represent user-item interactions through latent features and identify hidden user preferences.

RECOMMENDER SYSTEM

Collaborative filtering and machine-learning techniques are combined to generate personalized product recommendations based mainly on users' historical interactions and preferences.

PROCESSED DATA

The collected data is cleaned, analyzed, and transformed into suitable user-item information for building the recommendation model.

COLLABORATIVE APPROACH

The system identifies similar users and similar products from their interaction patterns and applies collaborative filtering to generate recommendations.

TOP-N RECOMMENDATION

A ranking mechanism selects the Top-N products for each user. The recommendation performance is evaluated using metrics such as Precision, Recall, and F1-score.

Proposed System

BEHAVIOR DATA

The system analyzes visitor ID, product ID, event type, and timestamp from the RetailRocket dataset to understand customer interactions.

PURCHASE PREDICTION

Logistic Regression, Decision Tree, and Random Forest are applied and compared to predict the customer's likelihood of purchasing a product.

PURCHASE JOURNEY

The system identifies the customer's current stage:

● Exploring → ● Considering → ● Ready to Purchase

RECOMMENDATIONS ARE DYNAMICALLY CHANGED ACCORDING TO THE DETECTED STAGE.

BEHAVIORAL FEATURES

Interaction data is transformed into features such as view frequency, cart activity, transaction history, and recent interactions.

CUSTOMER SEGMENTATION

K-Means Clustering groups customers with similar shopping behavior and identifies different customer patterns

PERSONALIZED RECOMMENDATION

The system generates recommendations based on customer behavior, predicted purchase probability, customer segment, and current journey stage, providing more relevant products.

Proposed System

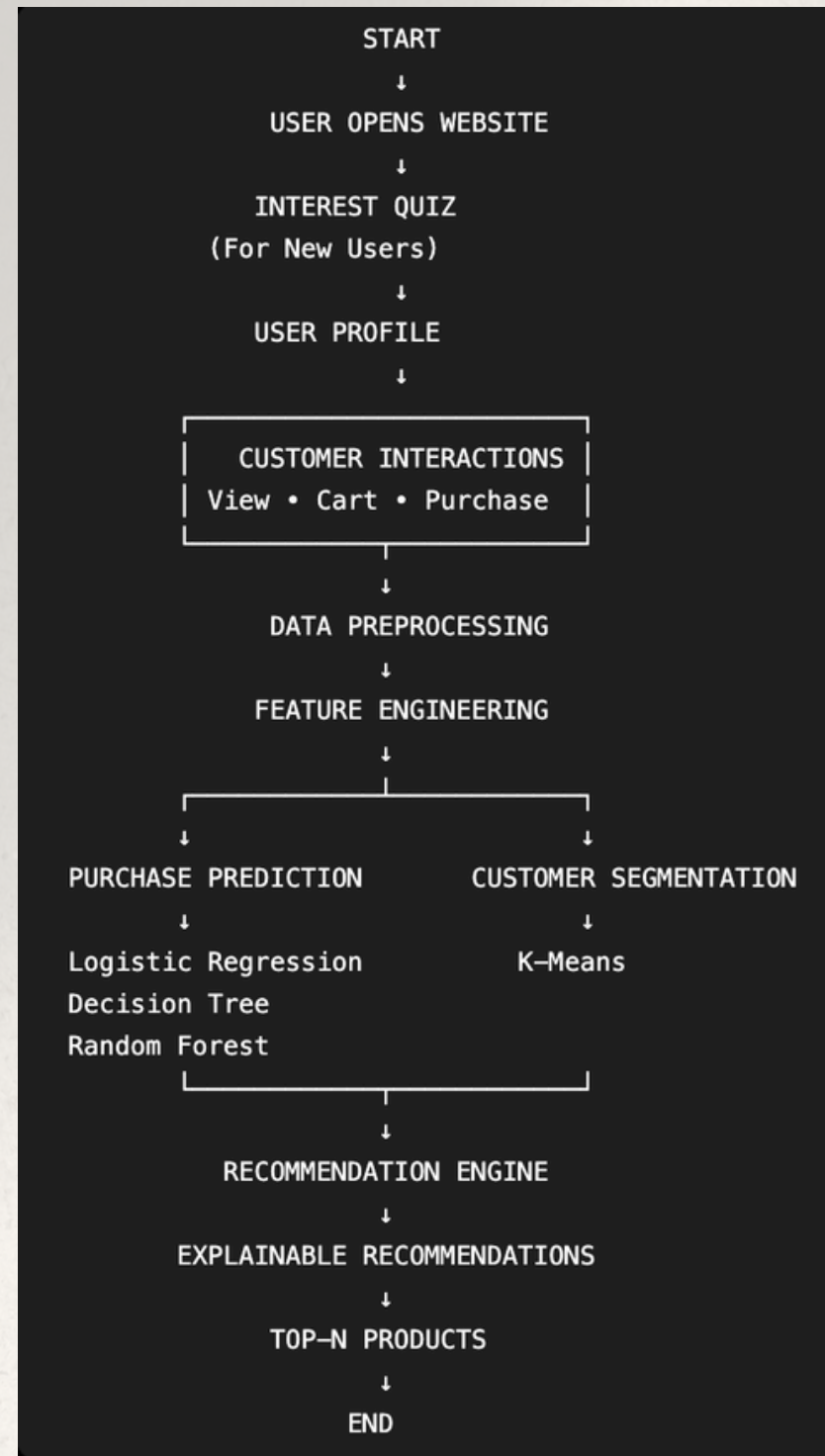
01 | Intelligent Customer Analysis

Our system analyzes customer-product interactions such as views, add-to-cart actions, and transactions. Behavioral features are extracted and used with Logistic Regression, Decision Tree, Random Forest, and K-Means to predict purchase behavior and identify customer segments.

02 | Personalized & Explainable Recommendations

Unlike a conventional recommendation system, our approach combines customer interests, observed behavior, and purchase patterns to generate relevant products. Each recommendation can also provide a simple “Why this?” explanation, helping users understand why a product was suggested.

Flow chart



1. USER OPENS THE WEBSITE → INTEREST QUIZ

FOR A NEW CUSTOMER, WE FIRST ASK WHAT THEY'RE INTERESTED IN. THESE ANSWERS CREATE THEIR INITIAL PREFERENCE PROFILE.

2. CUSTOMER INTERACTIONS

AS THE CUSTOMER USES THE WEBSITE, WE COLLECT THEIR ACTUAL BEHAVIOR — WHAT THEY VIEW, ADD TO CART, AND PURCHASE.

3. DATA PREPROCESSING → FEATURE ENGINEERING

WE CLEAN THE DATA AND CONVERT THESE RAW INTERACTIONS INTO USEFUL FEATURES SUCH AS NUMBER OF VIEWS, CART ADDITIONS, PURCHASES, INTERACTION FREQUENCY, AND PURCHASE RATE.

4. PURCHASE PREDICTION

WE USE LOGISTIC REGRESSION, DECISION TREE, AND RANDOM FOREST TO PREDICT WHETHER A CUSTOMER IS LIKELY TO PURCHASE.

5. CUSTOMER SEGMENTATION

AT THE SAME TIME, K-MEANS GROUPS CUSTOMERS WHO HAVE SIMILAR SHOPPING BEHAVIOR.

6. RECOMMENDATION ENGINE

THE SYSTEM COMBINES THE CUSTOMER'S PROFILE, BEHAVIOR, PREDICTED PURCHASE BEHAVIOR, AND CUSTOMER SEGMENT TO DECIDE WHICH PRODUCTS ARE RELEVANT.

7. EXPLAINABLE RECOMMENDATION

WE DON'T JUST SHOW A PRODUCT — WE CAN ALSO TELL THE CUSTOMER WHY IT WAS RECOMMENDED,

Algorithms

01 — LOGISTIC REGRESSION

Purchase Prediction

Predicts the probability of a customer making a purchase using behavioral features such as views, cart additions, and interaction frequency.

02 — HISTGRADIENTBOOSTING

Purchase Prediction — Final Model

Builds multiple decision trees step-by-step, where each new tree improves the mistakes of previous trees. It can capture complex customer behavior and interactions.

ROC-AUC-Measures how well the model separates likely purchases from non-purchases across different decision thresholds.

PR-AUC-Measures how well the model identifies the less-common purchase cases, balancing precision and recall.

SDG Goals

SDG 12 – Responsible Consumption & Production

- **Helps customers discover relevant products based on their interests and behavior.**
- **Supports more informed and efficient product discovery.**



SDG 9 – Industry, Innovation & Infrastructure

- **Uses Machine Learning to create an intelligent e-commerce system.**
- **Improves digital innovation through purchase prediction and personalized recommendations.**



Expected output

01 — PURCHASE PREDICTION

Predict whether a customer is likely to purchase a product.

02 — CUSTOMER SEGMENTATION

Group customers with similar shopping behavior.

03 — PERSONALISED RECOMMENDATIONS

Suggest products based on each customer's interests and behavior.

04—EXPLAINABLE RECOMMENDATIONS

Show why a product was recommended to the customer.

05 — MODEL COMPARISON

Identify the best-performing model using standard evaluation metrics.

06 — INTERACTIVE WEB PROTOTYPE

Display predictions, customer segments, and recommendations through a simple Streamlit interface.

Thank You